

Complete RAG (Retrieval-Augmented Generation) Syllabus

1. Foundations (Prerequisites)

Python Programming

- Functions
- OOP
- File handling
- Exception handling
- Decorators
- Iterators / generators
- Virtual environments
- Package management (pip, poetry)
- Async programming
- Libraries: numpy, pandas, requests, asyncio, pathlib, typing, pydantic

NLP Fundamentals

- Tokenization
- Stemming
- Lemmatization
- Stopword removal
- Text normalization
- N-grams
- TF-IDF
- Bag of Words
- Named Entity Recognition
- POS tagging
- Dependency parsing
- Semantic similarity
- BM25
- Precision / Recall / F1 / MAP / MRR / NDCG

Deep Learning Basics

- Neural networks
- Backpropagation
- Gradient descent
- CNN
- RNN
- LSTM
- Attention mechanism

2. Transformers & LLM Foundations

Transformer Architecture

- Encoder / Decoder
- Self-attention
- Cross-attention
- Query / Key / Value
- Positional encoding
- Layer normalization
- Residual connections
- Feedforward layers
- Masked attention
- Multi-head attention

Embeddings

- Word2Vec
- GloVe
- FastText
- Sentence Transformers
- Dense embeddings
- Sparse embeddings
- Hybrid embeddings
- Cosine similarity
- Dot product
- Euclidean distance

LLM Concepts

- Autoregressive models
- Masked language models
- Tokenization
- BPE / WordPiece / SentencePiece
- Context window
- Hallucination
- Prompting
- Temperature / Top-k / Top-p
- Fine-tuning
- Instruction tuning
- RLHF
- Quantization
- LoRA
- PEFT

3. Information Retrieval

- Document retrieval
- Passage retrieval
- Dense retrieval
- Sparse retrieval

- Hybrid retrieval
- Ranking / reranking
- TF-IDF
- BM25
- DPR
- Contriever
- ColBERT
- SPLADE
- Inverted indexes
- Posting lists
- ANN search
- FAISS
- IVF
- PQ
- HNSW
- ScaNN
- Annoy

4. RAG Core Architecture

- What is RAG
- Why RAG
- Hallucination reduction
- Knowledge grounding
- External memory

Types of RAG

- Basic RAG
- Naive RAG
- Advanced RAG
- Hybrid RAG
- Agentic RAG
- Graph RAG
- Multi-hop RAG
- Adaptive RAG
- Corrective RAG (CRAG)
- Self-RAG
- Fusion RAG
- Modular RAG
- Hierarchical RAG
- Long-context RAG

5. Document Processing Pipeline

- PDF ingestion
- DOCX ingestion
- HTML parsing
- APIs
- Databases
- OCR cleanup

- Duplicate removal
- Metadata extraction

Chunking

- Fixed chunking
- Recursive chunking
- Semantic chunking
- Sentence chunking
- Paragraph chunking
- Sliding window chunking
- Hierarchical chunking
- Token-aware chunking
- Adaptive chunking

6. Embedding Systems

- Embedding model selection
- Benchmarking
- Multilingual embeddings
- Domain-specific embeddings
- OpenAI embeddings
- BGE
- E5
- Instructor
- Sentence Transformers
- Cohere embeddings

7. Vector Databases

- FAISS
- Chroma
- Pinecone
- Weaviate
- Milvus
- Qdrant
- Elasticsearch
- OpenSearch

Concepts:

- Indexing
- Metadata filtering
- Persistence
- Sharding
- Replication
- Namespaces
- Collections
- Hybrid querying

8. Query Processing

- Query understanding
- Query rewriting
- Query decomposition
- Query expansion
- HyDE
- Query routing
- Multi-query retrieval
- Intent detection
- Ambiguity resolution

9. Retrieval Improvement

- Reranking
- Cross encoders
- Bi-encoders
- Late interaction retrieval
- Contextual compression
- Relevance filtering
- Top-k tuning
- Score thresholding
- Cohere rerank
- BGE reranker

10. Prompt Engineering for RAG

- Context injection
- Citation prompting
- Instruction design
- Grounding prompts
- Fallback prompts
- QA prompts
- Summarization prompts
- Extraction prompts

11. Generation Control

- Hallucination mitigation
- Constrained generation
- Response validation
- Answer verification
- Confidence estimation
- Refusal handling

12. Advanced RAG Architectures

Graph RAG

- Knowledge graphs

- Nodes / edges
- Entity extraction
- Graph traversal
- Neo4j

Agentic RAG

- Planning
- Tool use
- Iterative retrieval
- Reflection
- Memory
- LangGraph
- CrewAI
- AutoGen

Multi-hop RAG

- Iterative reasoning
- Chained retrieval

Long-context RAG

- Context packing
- Compression
- Token budgeting
- Lost-in-the-middle problem

13. Evaluation

Retrieval Metrics

- Recall@k
- Precision@k
- Hit rate
- MRR
- MAP
- NDCG

Generation Metrics

- BLEU
- ROUGE
- BERTScore
- Factual consistency

RAG Evaluation

- Faithfulness
- Groundedness
- Context precision
- Answer relevancy

Tools:

- RAGAS
- TruLens
- DeepEval
- LangSmith

14. Production RAG Engineering

- Caching
- Batching
- Async retrieval
- Latency optimization
- Throughput optimization
- Retries
- Fallbacks
- Circuit breakers
- Docker
- Kubernetes
- FastAPI
- REST APIs
- gRPC
- Logging
- Monitoring
- Tracing

15. Security & Safety

- Prompt injection
- Retrieval poisoning
- Data leakage
- Access control
- Authentication
- RBAC
- PII masking
- Secrets management

16. Frameworks & Tools

- LangChain
- LlamaIndex
- Haystack
- DSPy
- vLLM
- Ollama
- TGI
- LM Studio
- OpenAI API
- Anthropic API
- Cohere API
- Hugging Face

17. Multimodal RAG

- Image embeddings
- OCR
- Table retrieval
- PDF vision parsing
- CLIP
- Donut
- LayoutLM
- Florence
- Multimodal LLMs

18. Domain-Specific RAG

- Legal RAG
- Healthcare RAG
- Finance RAG
- Enterprise RAG
- Code RAG
- Customer support RAG

19. Research Papers

- Original RAG
- DPR
- ColBERT
- HyDE
- Self-RAG
- CRAG
- RAPTOR
- GraphRAG
- FiD
- Atlas

20. Hands-on Projects

Beginner

- PDF chatbot
- Document QA bot
- FAQ bot

Intermediate

- Hybrid search chatbot
- Reranking RAG
- Multi-document QA

Advanced

- Graph RAG assistant

- Agentic RAG researcher
- Enterprise knowledge assistant
- Multimodal RAG chatbot

21. System Design

- Scalable RAG architecture
- Online vs offline indexing
- Streaming ingestion
- Distributed vector DB
- Cold start
- Multi-tenant architecture

22. Interview Topics

- Explain RAG architecture
- Dense vs sparse retrieval
- Chunking strategies
- ANN algorithms
- Vector DB comparisons
- Hallucination handling
- Reranking
- Hybrid retrieval
- Evaluation metrics
- Production bottlenecks

Learning Order

1. Python
2. NLP basics
3. Transformers
4. Embeddings
5. Retrieval systems
6. Vector databases
7. Chunking
8. Basic RAG
9. Hybrid retrieval
10. Reranking
11. Evaluation
12. Production RAG
13. Advanced RAG
14. Agentic RAG
15. Multimodal RAG